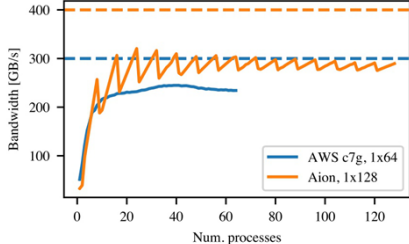
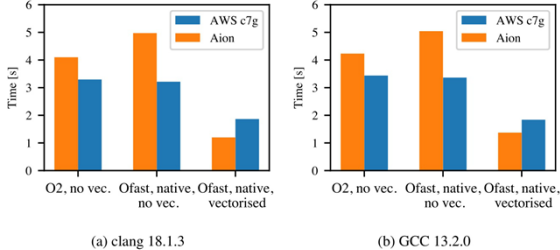
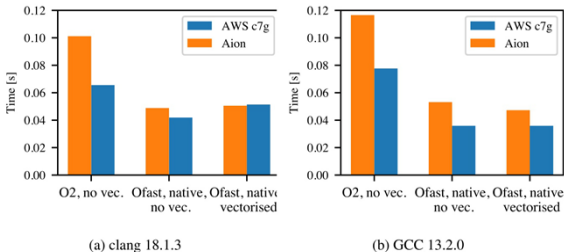
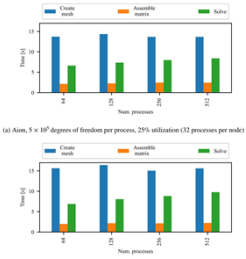


Alternative Texts for Your Images, Please Check and Correct them if Required

Page no	Fig/Photo	Thumbnail	Alt-text Description
4	fig1.jpg	 A line chart showing bandwidth in GB/s on the y-axis (0 to 400) against the number of processes on the x-axis (0 to 120). Two lines are plotted: a blue line for 'AWS c7g, 1x64' and an orange line for 'Aion, 1x128'. The blue line starts at approximately 100 GB/s at 0 processes, rises to about 250 GB/s at 40 processes, and then levels off. The orange line starts at approximately 100 GB/s at 0 processes, rises to about 300 GB/s at 40 processes, and then fluctuates between 280 and 320 GB/s. Two horizontal dashed lines are present: a blue one at approximately 300 GB/s and an orange one at approximately 400 GB/s.	Line chart showing bandwidth in GB/s versus the number of processes. Two lines represent different configurations: "AWS c7g, 1x64" in blue and "Aion, 1x128" in orange. The orange line shows higher variability and peaks compared to the blue line. Horizontal dashed lines indicate reference bandwidth levels at approximately 300 GB/s and 400 GB/s.
7	fig2.jpg	 Two bar charts comparing execution times in seconds. The left chart is labeled '(a) clang 18.1.3' and the right chart is labeled '(b) GCC 13.2.0'. Both charts have 'Time [s]' on the y-axis (0 to 6) and three categories on the x-axis: 'O2, no vec.', 'Ofast, native, no vec.', and 'Ofast, native, vectorised'. Each category has two bars: a blue bar for 'AWS c7g' and an orange bar for 'Aion'. In both charts, the orange bars are generally higher than the blue bars. The legend indicates AWS c7g in blue and Aion in orange.	Bar charts comparing execution times in seconds for AWS c7g and Aion across different compiler settings. The left chart shows results for clang 18.1.3, and the right chart for GCC 13.2.0. Each chart includes three configurations: "O2, no vec.", "Ofast, native, no vec.", and "Ofast, native, vectorised." The legend indicates AWS c7g in blue and Aion in orange.
7	fig1.jpg	 Two bar charts comparing execution times in seconds. The left chart is labeled '(a) clang 18.1.3' and the right chart is labeled '(b) GCC 13.2.0'. Both charts have 'Time [s]' on the y-axis (0.00 to 0.12) and three categories on the x-axis: 'O2, no vec.', 'Ofast, native, no vec.', and 'Ofast, native, vectorised'. Each category has two bars: a blue bar for 'AWS c7g' and an orange bar for 'Aion'. In both charts, the orange bars are generally higher than the blue bars. The legend indicates AWS c7g in blue and Aion in orange.	Two bar charts comparing execution times for different compiler settings. The left chart, labeled "clang 18.1.3," and the right chart, labeled "GCC 13.2.0," both display time in seconds on the y-axis. The x-axis categories are "O2, no vec.," "Ofast, native, no vec.," and "Ofast, native, vectorised." Each category has two bars: blue for "AWS c7g" and orange for "Aion." The charts show variations in execution times across different settings and compilers.

Page no	Fig/Photo	Thumbnail	Alt-text Description
10	fig1.jpg	 (a) Aion, 5×10^5 degrees of freedom per process, 25% utilization (32 processes per node)	Bar chart showing the time in seconds for three tasks: creating a mesh, assembling a matrix, and solving, across different numbers of processes (64, 128, 256, 512). The chart indicates that creating a mesh takes the most time, followed by solving, and then assembling a matrix. The data is for Aion with 5×10^5 degrees of freedom per process and 25% utilization, with 32 processes per node.